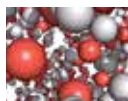




NASA unveils space helicopter

NASA has unveiled the first-ever 'mini' helicopter to fly around another planet, namely Mars', atmosphere. <https://digit.in/jun18-B29-1>



World's fastest water heater

Scientists use an X-Ray laser to heat water to 100,000 degrees Celsius in less than tenth of a picosecond. <https://digit.in/jun18-B29-2a>

THE NEXT DECADE OF SPACE EXPLORATION

We will see a number of new launch vehicles, study the Red Planet to prepare for pioneering manned missions, and learn about the icy moons in unprecedented detail

The
Parker
Solar
Probe

Aditya Madanapalle |

aditya@digit.in



One of the most exciting launches scheduled is that of the James Webb Space Telescope (JWST). This next generation telescope is so powerful, that it can detect a bumblebee on the surface of the moon. It will be used to observe exoplanets in greater detail and image the universe in more wavelengths than the HST. It will be able to look into the earliest galaxies to form in the Universe. Expect bigger and more awe-inspiring images and science than what the HST gave us. The Webb Telescope is scheduled for launch in 2020.

One of the biggest planned missions for ISRO is its first manned spaceflight. A fully autonomous orbital capsule will house a crew of two or three "brahmanauts". They will remain in orbit for between seven to

twelve days. The rocket used for the spaceflight will be the Geosynchronous Satellite Launch Vehicle, also known as "fatboy" or "Baahubali". ISRO will also be developing its Reusable Launch Vehicle, or the Indian version of the Space Shuttle.

Constellations of navigation, communication and Earth Observation Satellites will be deployed. At least two companies, SpaceX and OneWeb are both planning constellations dedicated to provide worldwide broadband services. The SpaceX constellation, called Starlink will have 12,000 satellites, while the OneWeb constellation is made up of 648 satellites.

Expect more countries to start up their own space programs, and we will also see the first space tourists. SpaceX, Lockheed Martin, Boeing and Blue Origin are all in the space tourism race. Bigelow Aerospace is expected to construct and deploy the first space hotel in the 2020s. These are likely to be formed of a chain of inflatable modules. The ISS is scheduled to be decommissioned

after another ten years, but the Chinese space station is expected to be in orbit. In a mission that appears to be straight out of a space exploration video game. The Prospector 1 by Deep Space Industries is expected to be the first to evaluate a near Earth asteroid for its worth. That is, to figure out how viable private asteroid mining really is.

LAUNCH VEHICLES

Launch vehicles are the bread and butter of spaceflight. It does not matter if it is a novel probe to the outer planets, a pioneering manned mission, or just another communication satellite in orbit. Rockets are required to take stuff up into space. Right now, there are just not enough satellite launch systems to serve the demand. Over the course of the next decade though, we will be seeing a number of rockets come up, both big and small. Around 2019, the first flight of the next generation of NASA launch vehicles should take off. This is the Space Launch System, which will provide